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(54) **ALL-SOLID-STATE ALUMINUM-ION
BATTERY**

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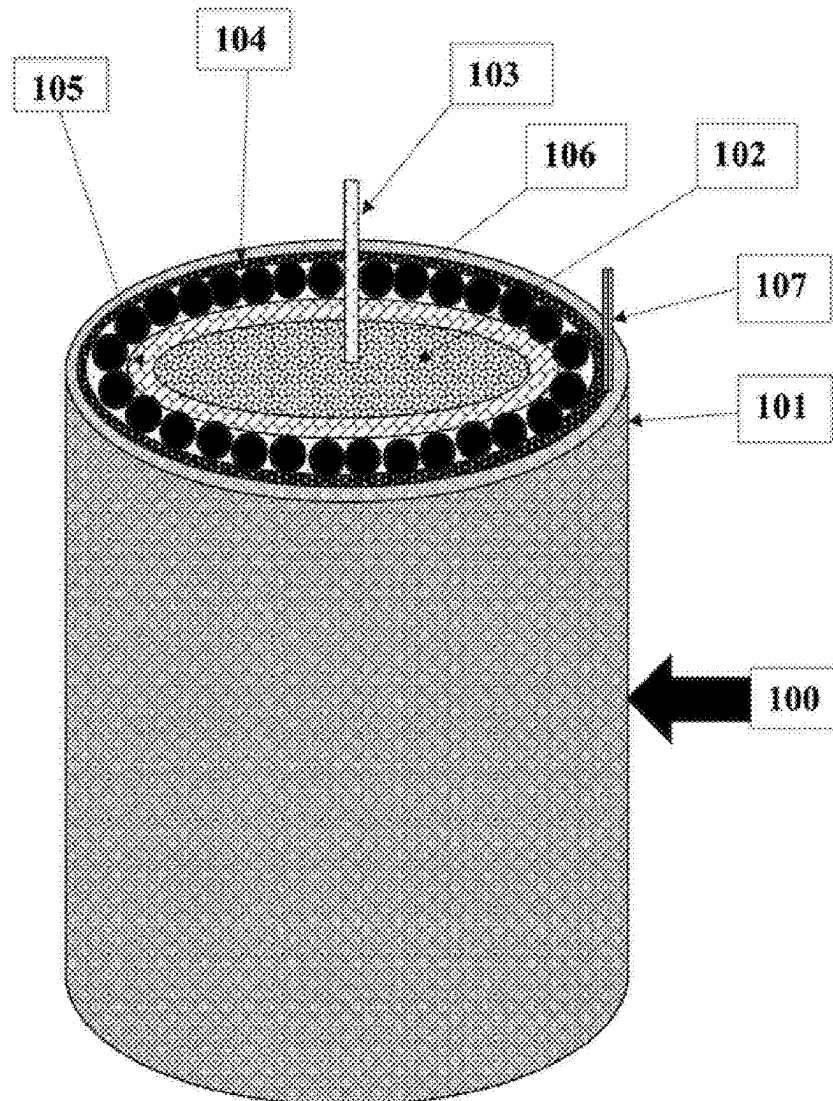
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ABSTRACT

A method and apparatus for storing electricity using a rechargeable electrochemical cell with axially placed coiled aluminum wire anode, peripherally bunched carbon graphite rods as the cathode, and a solid electrolyte. The invention teaches the use of common ultra-low-cost material components, and a simple cell construction method.



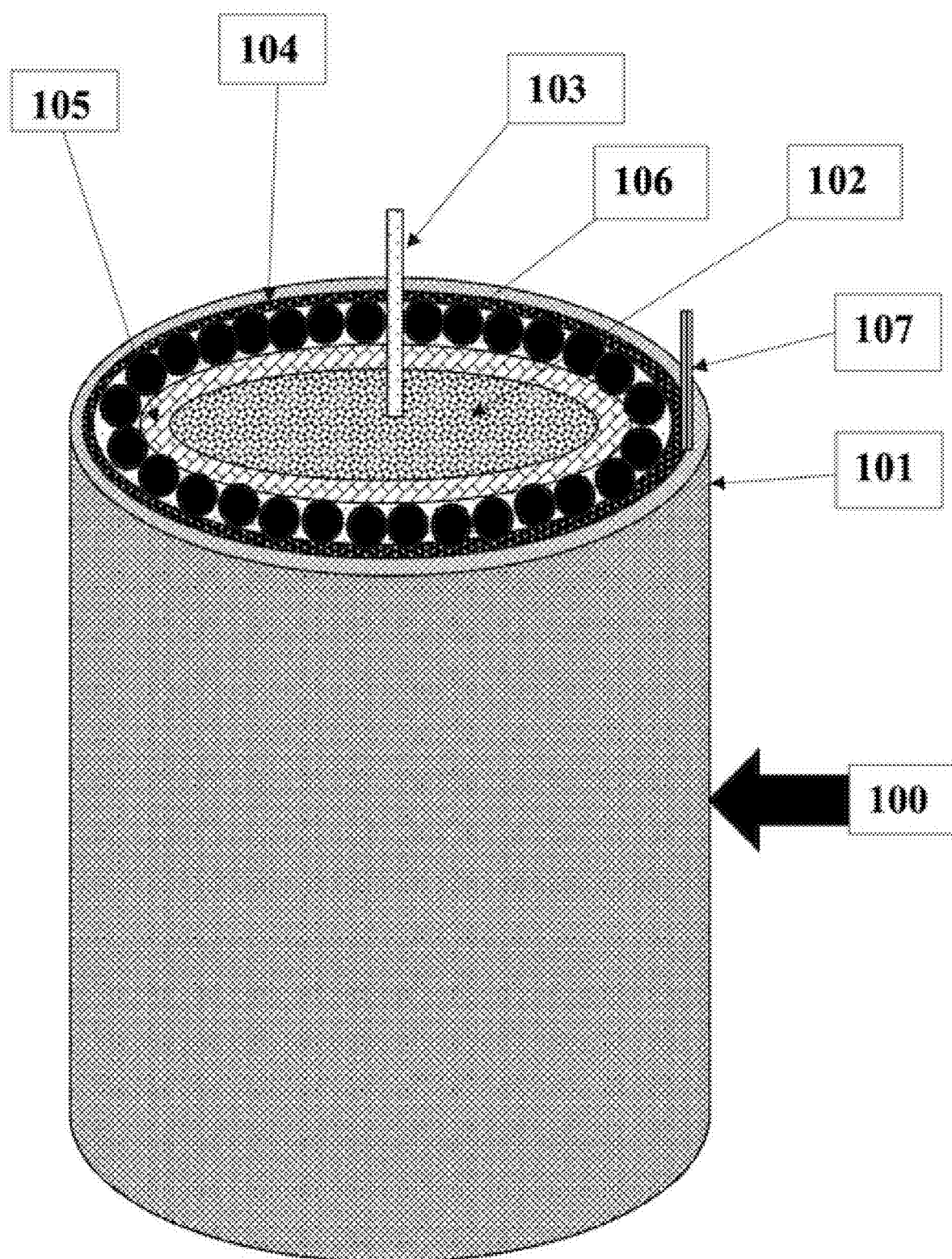


Fig. 1

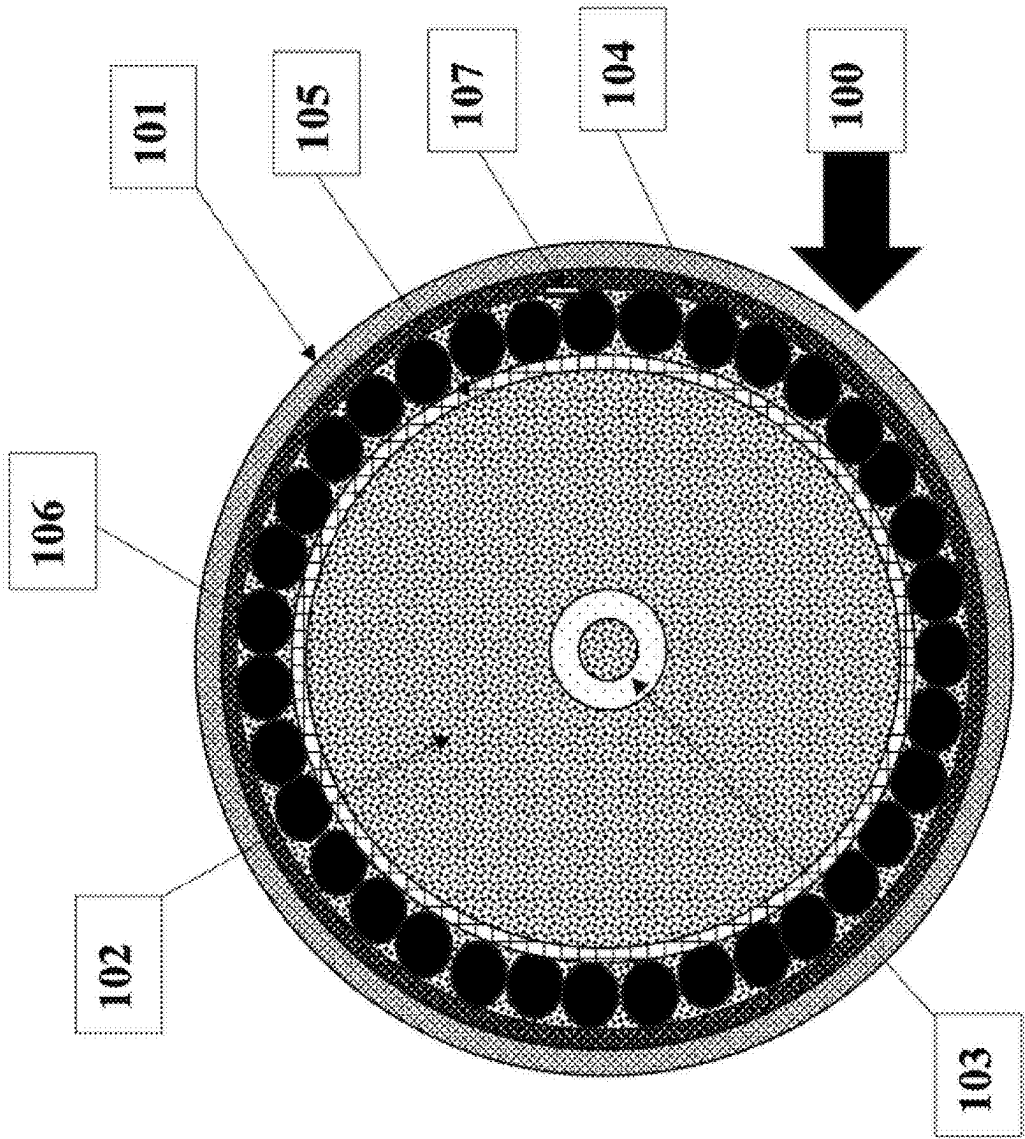


Fig. 2

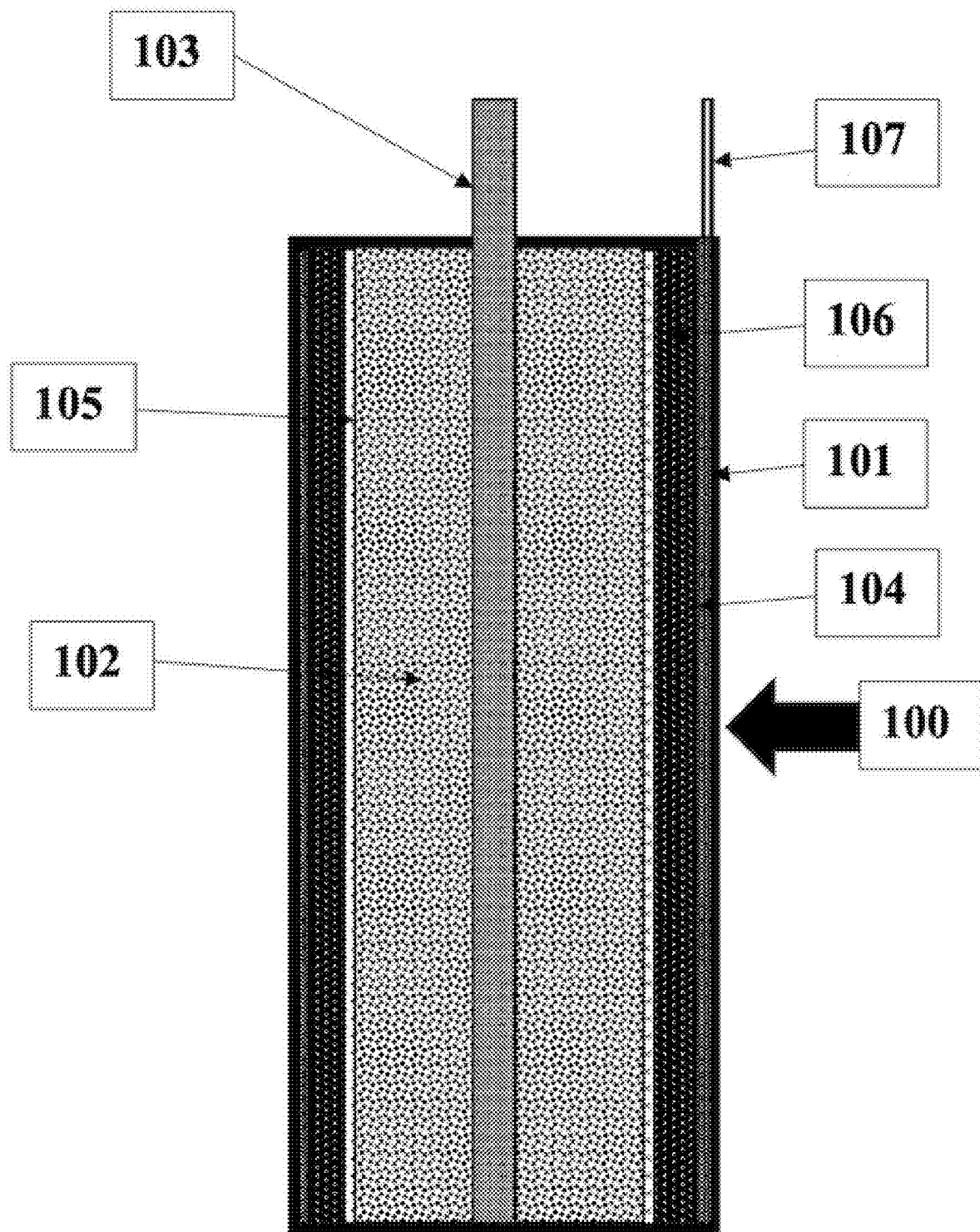


Fig. 3

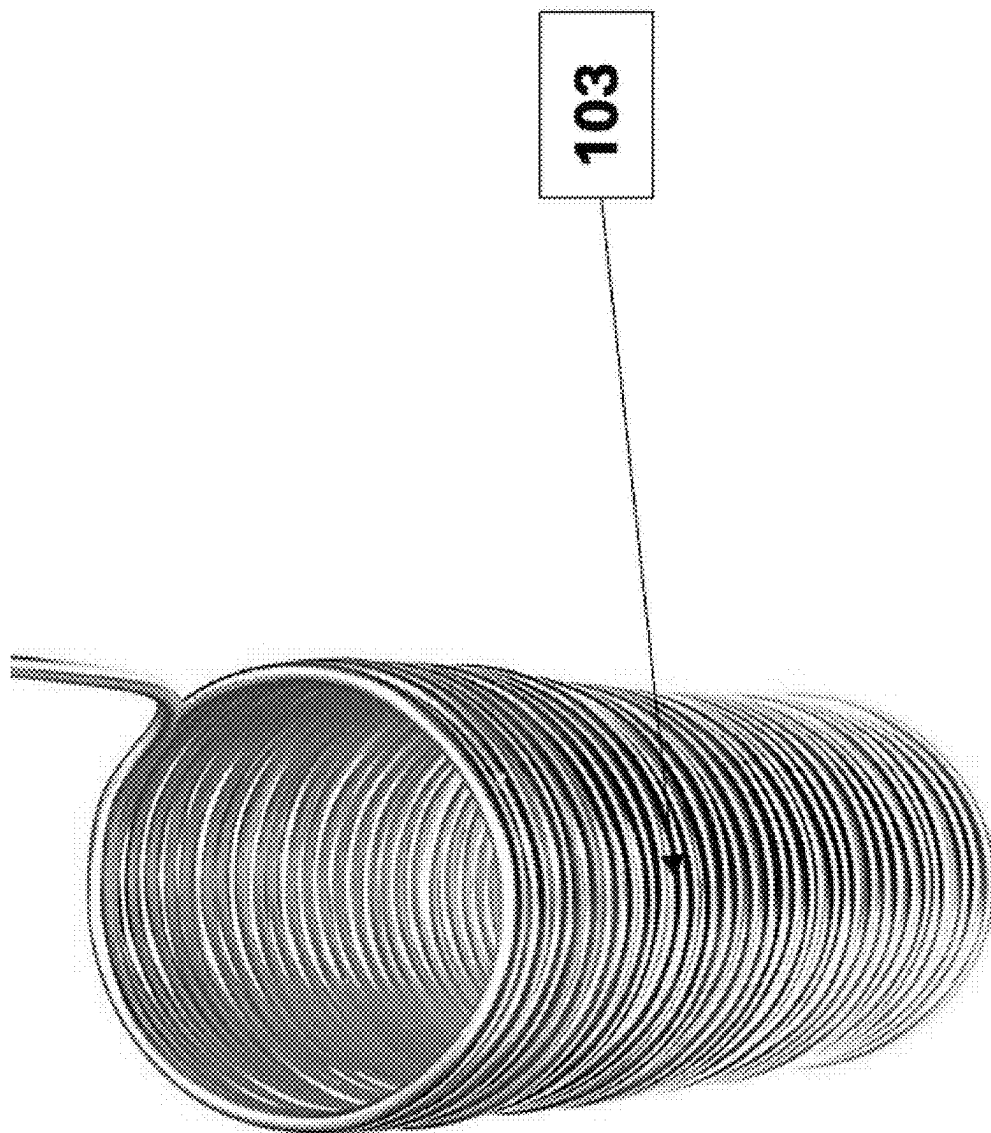
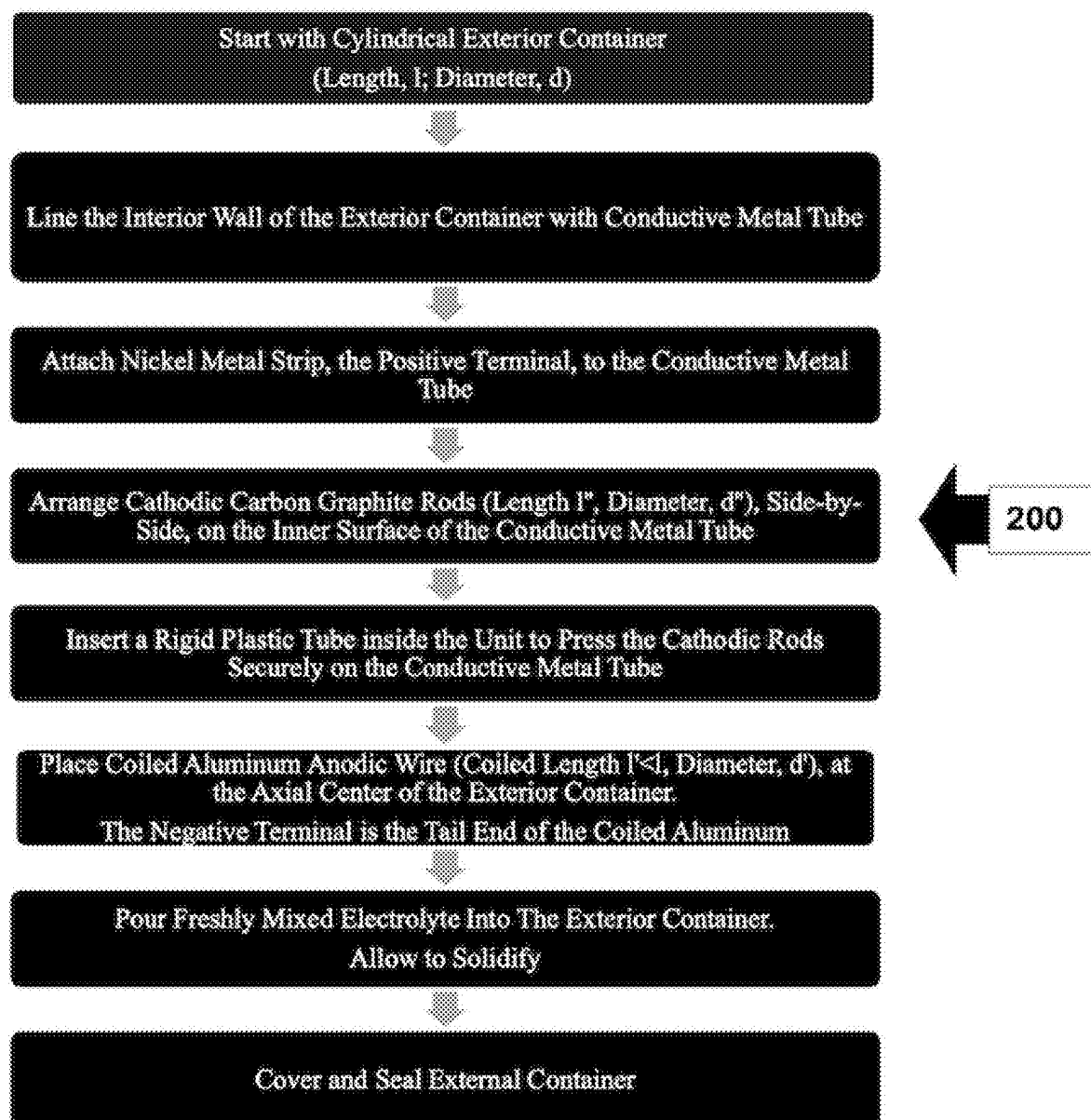


Fig. 4

*Fig. 5*

ALL-SOLID-STATE ALUMINUM-ION BATTERY

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FIELD OF INVENTION

[0033] The present invention generally relates to electricity generation, and more particularly to a method and apparatus for electrical energy storage using rechargeable aluminum-ion electrochemical cells.

PRIOR ART

[0034] An aluminum-ion battery is a rechargeable battery in which aluminum ions serve as charge carriers. Aluminum-ion batteries consist of electrodes immersed in an electrolyte. The obvious material for the anode is aluminum, an alloy of aluminum, or a compound of aluminum. Common materials for the cathode (Pan, et al., 2022) include: carbon-based materials such as graphite, amorphous carbons, porous carbons (Li, et al., 2018). Common materials for the electrolyte include $\text{CO}(\text{NH}_2)_2$ or carbamide (also called urea), sea-salt, and acidic room temperature non-aqueous ionic liquids (IL). The ionic liquid is made of aluminum chloride (AlCl_3) and 1-ethyl-3-methylimidazolium chloride $[\text{EmIm}]\text{Cl}$ (Miguel, et al., 2020). The use of the ionic liquid as an electrolyte prevents passivation.

[0035] Aluminum-ion batteries are promising alternatives to the lithium-ion batteries commonly used in portable electronics, electric vehicles, and stationary energy for homes, commercial buildings, and grid-scale applications. Advantages of aluminum-ion batteries include the relatively low cost, energy density, safety, and long cycle life (Leise-gang, et al., 2019).

[0036] Recent investigations on aluminum-ion batteries have been conducted by:

[0037] (Wang, et al., 2017) who presented an advanced rechargeable aluminum-ion battery with a high-quality natural graphite cathode.

[0038] (Das, et al., 2017) who detailed developments and challenges faced by aluminum-ion batteries. The authors focus on electrode materials, innovative perspectives, and future research efforts on rechargeable aluminum-ion batteries.

[0039] (Gan, et al., 2019) who designed a high-performance rechargeable aluminum battery using a graphite cathode and $\text{AlCl}_3/\text{Et}_3\text{NHCl}$ ionic liquid electrolyte.

[0040] (Levy & Ein-Eli, 2020) who discussed the potential of aluminum-ion batteries as a cost-effective energy storage technology for renewable energy sources.

[0041] (Craig, et al., 2020) who reviewed current progress in non-aqueous aluminum batteries. The authors conclude that graphite-based positive electrodes, especially cheap and abundant graphite flakes, offer the best overall performance because of the stability and high discharge potential.

[0042] (Elia, et al., 2021) who reviewed developments in different classes of aluminum-centered batteries. The focus was on “aluminum electrolyte chemistry based on chloroaluminate melts, deep eutectic solvents, polymers, and chlorine-free formulations.”

[0043] (Yuan, et al., 2020) which describes the status, challenges, emerging, and outlooks of emerging rechargeable aqueous aluminum-ion battery.

[0044] Patents which taught implementations of the aluminum-ion batteries include:

[0045] (Archer, et al., 2014) which teaches an aluminum ion battery using an aluminum anode, a vanadium oxide material cathode, and an ionic liquid electrolyte. The vanadium oxide material cathode comprises a monocrystalline orthorhombic vanadium oxide material.

[0046] (Mukherjee & Koratkar, 2017) which describes a rechargeable battery using a solution of aluminum salt as an electrolyte.

[0047] (Huang & Chen, 2018) which describes an aluminum-ion battery using an electrolyte comprising of an aluminum halide, a solvent and a compound made from alkyl or fluoroalkyl group.

[0048] (Gao & Chen, 2018) which discloses a method of preparing a graphene anode material by coating a graphene oxide solution on a substrate, drying, removing the substrate, performing reduction, and obtaining a graphene film with ultra-high conductivity.

[0049] (Mukherjee, et al., 2021) which describes an aqueous aluminum ion battery with improved charge storage capacity.

[0050] (Stoddart, et al., 2021) which discloses rechargeable aluminum batteries using cathodic phenanthrene-quinone unit and a graphite flake.

[0051] (Azimi & Ng, 2023) which describes aluminum-ion battery technology with an electrolyte consisting of an aluminum trichloride ($\text{Al}-\text{Cl}_3$)/trimethylamine hydrochloride ionic liquid, aluminum metal as the anode material, and a compatible cathode active material.

[0052] (Su, et al., 2020) which describes an aluminum-ion battery with a cathode comprising of a layer of recompressed exfoliated graphite or carbon material that is oriented in such a manner that the layer has a graphite edge plane in direct contact with the electrolyte and facing the separator.

[0053] (Caban-Acevedo & Yushin, 2023) which describes rechargeable batteries with alkali metal ion cathodes, aluminum metal-based anodes and displacement electrolyte.

BACKGROUND OF THE INVENTION

[0054] Aluminum-ion batteries are emerging as better alternatives to lithium-ion batteries, the leading choice for wireless devices, computers, electric mobility, and small-scale to grid-scale stationary energy storage applications. The advantages of aluminum-ion batteries over lithium-ion batteries include: cost-effectiveness of aluminum because of its abundance in the Earth's crust; safety because aluminum-ion batteries are less prone to thermal runaway and fire hazards; significantly higher theoretical energy density, 1060 Wh/kg versus 406 Wh/kg theoretical limit for lithium-ion batteries; high recyclability of aluminum; longevity due to the large number of charging and discharging cycles of aluminum-ion batteries.

[0055] This invention uses the advantages inherent in the electrochemistry of aluminum-ion batteries to teach the construction of an ultra-low-cost all-solid-state rechargeable battery. The electrodes maximize electrochemical reaction surfaces. The cells are easy to build. Cell components are affordable and widely available materials.

SUMMARY OF THE INVENTION

[0056] According to the present invention there is provided a method of storing electrical energy using a membrane-free electrochemical cell comprising a cylindrical exterior container, a coiled anodic aluminum wire, bunched carbon graphite cathodic rods, and a solid electrolyte based on a compound mixture of urea, sea-salt, and sodium silicate.

[0057] An advantage of the present invention is the provision of a method and apparatus for converting electricity into chemical energy stored in an aluminum-ion cell.

[0058] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which utilizes a solid electrolyte.

[0059] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which utilizes a coiled anodic aluminum wire.

[0060] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which utilizes bunched cathodic carbon graphite rods.

[0061] Still another advantage of the present invention is the provision of a method and apparatus for electricity storage which is membrane-free.

[0062] Still other advantages of the invention will become apparent to those skilled in the art upon reading and understanding the following detailed description, accompanying drawings and appended claims.

BRIEF DESCRIPTION OF DRAWINGS

[0063] The invention may take physical form in certain parts and arrangements of parts, a preferred embodiment and method of which will be described in detail in this specification and illustrated in the accompanying drawings which form a part hereof, and wherein:

[0064] FIG. 1 is an illustration of the isometric view of the Aluminum-Ion Cell;

[0065] FIG. 2 is an illustration of the horizontal cross-section of the Aluminum-Ion Cell;

[0066] FIG. 3 is an illustration of the vertical cross-section of the Aluminum-Ion Cell;

[0067] FIG. 4 is an image of the coiled aluminum anode;

[0068] FIG. 5 is an illustration of the steps involved in the construction of the aluminum-ion cell with the coiled aluminum anode, the bunched peripheral cathodic carbon graphite rods, and the solid electrolyte.

DETAILED DESCRIPTION OF THE INVENTION

[0069] It should be appreciated that while a preferred embodiment of the present invention will be described with reference to aluminum-ion electrochemical cell, other metal-ion electrochemical cells are also suitable for use in connection with the present invention. These include zinc-ion, iron-ion, magnesium-ion, lithium-ion, calcium-ion, sodium-ion, potassium-ion, and tin-ion electrochemical cells.

[0070] In accordance with a preferred embodiment, the present invention teaches the storage of electricity using an electrochemical cell comprising a) a cylindrical exterior container; b) a coiled axially placed anodic wire; c) a peripherally bunched group of cathodic rods; and d) an electrolyte based on a mixture of urea, sea-salt, and sodium silicate.

[0071] Referring now to the drawings wherein the showings are for the purposes of illustrating a preferred embodiment of the invention only and not for purposes of limiting same.

[0072] Referring now to FIG. 1:

[0073] Electrochemical cell 100 has exterior container 101. Coiled anodic aluminum wire 103 is placed at the axial center of exterior container 101. Cathode 106

consists of carbon graphite rods arranged peripherally along the interior wall of exterior container 101. Solid electrolyte 102 occupies the entirety of the exterior container except the spaces taken up by anodic coil 103, and cathode 106.

[0074] Referring now to FIG. 2 and FIG. 4:

[0075] Anode 103. According to this invention, anode 103 is a cylindrical coil, made by spirally winding a single aluminum wire. The diameter of the single wire should not exceed 3 mm. The purity of the aluminum wire must be 99% at the minimum.

[0076] Referring now to FIG. 3:

[0077] 1. Exterior Container 101. According to this invention, exterior container 101 consists of polycarbonate rigid tubing of wall thickness not exceeding 1 mm.

[0078] 2. Electrolyte 102. According to this invention, electrolyte 102 is formulated by mixing x parts by volume of powdery urea $(\text{NH}_2)_2\text{CO}$ with y parts by volume of sea-salt, with z parts by volume of sodium silicate (water glass). The mixture is stirred until a uniformly smooth gel is achieved. Freshly mixed electrolyte 102, with special urea/sea-salt/water glass mix ratios, is characterized by rapid solidification. Therefore, it must be injected quickly into the cavity of electrochemical cell 100 prior to solidification.

[0079] 3. Cathode 103. According to the preferred embodiment of this patent, cathode 103 consists of carbon graphite rods, each of diameter not exceeding 6 mm, arranged side-by-side along the inside periphery of exterior container 101.

[0080] 4. Current Collector 104. According to the preferred embodiment of this patent, current collector 104 consists of a conductive metal tube, made preferably by folding a plain sheet, foil, or mesh of copper or stainless steel into a cylindrical shape. Current collector 104 is sandwiched between exterior container 101 and cathode 103.

[0081] 5. Rigid Mesh Tube 105: According to the preferred embodiment of this patent, rigid mesh tube 105 presses cathode 103 securely against current collector 104. Rigid mesh tube 105 is made of polypropylene plastic with thickness less than 2 mm and open-area that ranges between 37% and 48%.

[0082] 6. Positive Terminal 107: According to the preferred embodiment of this patent, positive terminal 107, is a thin nickel strip (thickness 0.15 mm, width 8 mm) attached to current collector 104.

[0083] Referring now to FIG. 5: The steps 200 involved in the construction of aluminum-ion cell 100 are:

[0084] 1. Start with Cylindrical Exterior Container 101 (Length, l; Diameter, d)

[0085] 2. Line the Interior Wall of the Exterior Container 101 with Conductive Metal 104.

[0086] 3. Attach Positive Terminal 107 to Conductive Metal 104.

[0087] 4. Arrange Cathodic Carbon Graphite Rods 103 (Length l", Diameter, d"), Side-by-Side, on the Inner Surface of the Conductive Metal 104.

[0088] 5. Insert Rigid Plastic Tube 105 inside the Unit to Securely Press the Cathodic Rods 106 on Conductive Metal 104.

[0089] 6. Place Coiled Wire Aluminum Anode **103** (Coiled Length $l' < l$, Diameter, d'), at the Axial Center of the Exterior Container **101**.

[0090] 7. Pour Freshly Mixed Electrolyte **102** into Exterior Container **101**. Allow to Solidify.

[0091] 8. Cover and Seal Exterior Container **101**.

EXAMPLE

[0092] To establish the characteristics of the aluminum-ion electrochemical cell, according to the invention disclosed herein, charge and discharge tests were conducted on a cell of distinct size configurations, anodic coil, and cathodic carbon graphite pack. The electrolyte mixture consists of 10 ml of urea, 10 ml of sea-salt, and 120 ml of sodium silicate.

[0093] Diameter of Exterior Container: 50 mm

[0094] Length of Exterior Container: 100 mm

[0095] Anode: Cylindrical coil (Diameter: 12 mm, Height: 70 mm) made by spirally winding 2 mm (Gauge 12) aluminum wire.

[0096] Cathode: 23 graphite rods (Diameter: 6 mm; Length: 90 mm) arranged side-by-side along the periphery of the exterior container.

[0097] Discharge current: 10 mA

[0098] Discharge test duration: 1,253 minutes

[0099] Current Capacity: 209 mAh

[0100] Energy Produced: 234 mWh

[0101] Open Circuit Voltage: 2.25V

[0102] Average Operating Voltage: 1.12V

[0103] The present invention has been described with reference to a preferred embodiment. Obviously, modifications and alterations will occur to others upon a reading and understanding of this specification. It is intended that all such modifications and alterations be included as far as they come within the scope of the appended claims or the equivalents thereof.

1. According to the present invention there is provided a method of storing electrical energy using an electrochemical cell comprising a cylindrical exterior container, coiled anodic wire, peripherally bunched cathodic rods, and electrolyte based on a compound mixture of urea, sea-salt, and sodium silicate.

2. A method according to claim 1, wherein the cathode consists of side-by-side carbon graphite rods packed tightly on the interior periphery of the exterior container.

3. A method according to claim 1, wherein the anode consists of a cylindrical aluminum coil placed at the axial center of the cylindrical exterior container.

4. A method according to claim 1, wherein the electrolyte is solid, consisting of a compound mixture of urea, sea-salt, and sodium silicate.

5. A method according to claim 2, wherein a conductive metal is placed between the cathode and the exterior container.

6. A method according to claim 5, wherein a rigid mesh tube securely presses the cathode against the conductive metal.

* * * * *